MATHEMATICS SOCIETY · ETON COLLEGE

RANDOM WALKS AND CONSENSUS

Random walks, random voters, and the dimension of space

Catalin Botezat · 21 October 2026



A WALK NEAR ZURICH

THE COUPLE IN THE WOOD

The mathematician George Pólya liked to walk in a wood near Zurich. On one walk he kept bumping into the same young couple, again and again. Was that just bad luck, or was it bound to happen?

TONIGHT'S QUESTION

If you wander at random forever, are you certain to come back?

THE PLAN

FOUR STEPS

01

THE WALKER

Will a random wanderer come home?

02

THE VOTERS

Will a crowd of voters end up agreeing?

03

THE TRICK

Why those are secretly the same question

04

WHY IT MATTERS

Family trees, ecosystems, networks and open problems

All you need: probability as a proportion, counting paths, and whether a sum is finite or infinite.

01

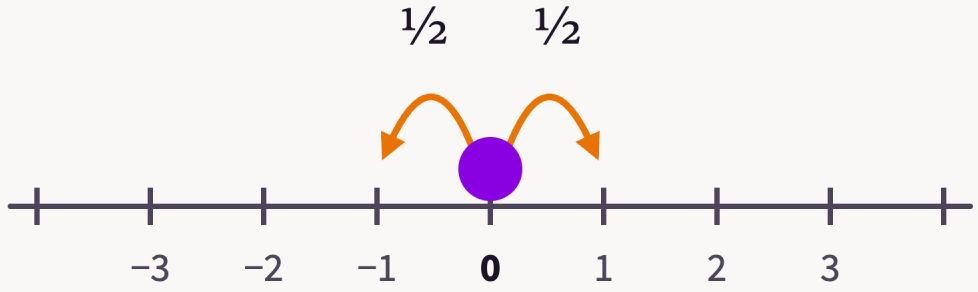
THE RANDOM WALKER

Will a random wanderer come home?

THE RULES

FLIP A COIN, TAKE A STEP

Heads: one step right. Tails: one step left. Repeat forever. "Home" is where you started.



MEMORYLESS

Where you go next depends only on where you are now, not on how you got there. Anything with this property is called a **Markov chain**.

> Andrei Markov invented these in 1913 by counting vowels and consonants in 20,000 letters of Pushkin's *Eugene Onegin*. Today they drive predictive text, weather models and Google.

VOTE 1 OF 4 · A LINE

A walker on a line. Will it come back to 0 with probability 1?

✓ Yes — certain

Hands up: no

VOTE 1 · THE ANSWER

YES, WITH CERTAINTY

Theory: exactly 99.4% are back by step 20,000, and 100% eventually. Orange dots are visits home.



Walkers back home at least once

99%

of 1,000 walkers, after 20,000 steps

WHY? COUNT THE PATHS

COUNTING PATHS HOME

To be home after 2n steps you need exactly n steps left and n steps right.

$$p_n = \binom{2n}{n} \Big/ 4^n$$

The top number, "2n choose n", counts the ways to choose which n of the 2n steps go left. There are $4^n = 2^{2n}$ paths in all, each equally likely.

Stirling's formula, for large n:

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

Steps	Paths home ÷ all	Chance	Estimate
2	2 ÷ 4	0.50	0.56
4	6 ÷ 16	0.38	0.40
6	20 ÷ 64	0.31	0.33
8	70 ÷ 256	0.27	0.28
10	252 ÷ 1,024	0.25	0.25

The chances shrink slowly. The estimate column is Stirling's formula:

$$\approx \frac{1}{\sqrt{\pi n}}$$

THE MOST IMPORTANT SLIDE OF PART 1

THE KEY PRINCIPLE

Count the average number of home visits per walker two different ways, then set the two counts equal.

COUNTED BY TIME

Run 1,000 walkers. At step 2n, a p_n fraction are home. Add that fraction over every time:

$$p_1 + p_2 + p_3 + \cdots$$

COUNTED BY ATTEMPT

Say the chance of returning at all is q. Memoryless, so chance of a 2nd, 3rd, … return is q^2, q^3, …:

$$q + q^2 + q^3 + \cdots = \frac{q}{1-q}$$

SAME AVERAGE, TWO ROUTES

$$p_1 + p_2 + p_3 + \cdots = \frac{q}{1-q}$$

q < 1: right side is finite, so the walker's total home visits is finite — it eventually leaves for good.
q = 1: right side is infinite, so the walker returns forever.

BACK TO THE LINE

1D: AN INFINITE TOTAL

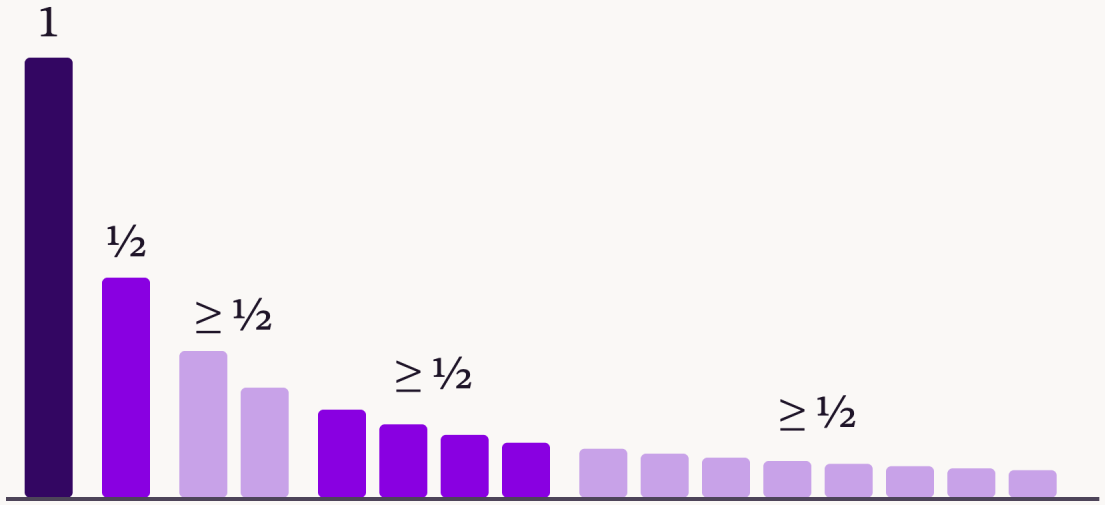
Each chance of being home is at least 1/(πn):

$$\frac{1}{\sqrt{\pi n}} \quad \geq \quad \frac{1}{\pi n}$$

so the total is at least

$$\frac{1}{\pi}\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots\right)$$

That is the harmonic series, and it is **infinite**. So the walker on a line comes home for certain.



Blocks of 1, 2, 4, 8, … terms each add up to at least ½, so the total grows without limit (Oresme, around 1350).

VOTE 2 OF 4 · A GRID

A walker on a grid. Will it come back home with probability 1?

Each step: north, south, east or west, with chance ¼ each.

✓ Yes — certain

Hands up: no

VOTE 2 · THE ANSWER

YES, STILL CERTAIN, BUT SLOWLY

Theory: exactly 79.0% are back by step 200,000; 99% takes about 10^{135} steps. Yet 100% return.



Walkers back home at least once

77%

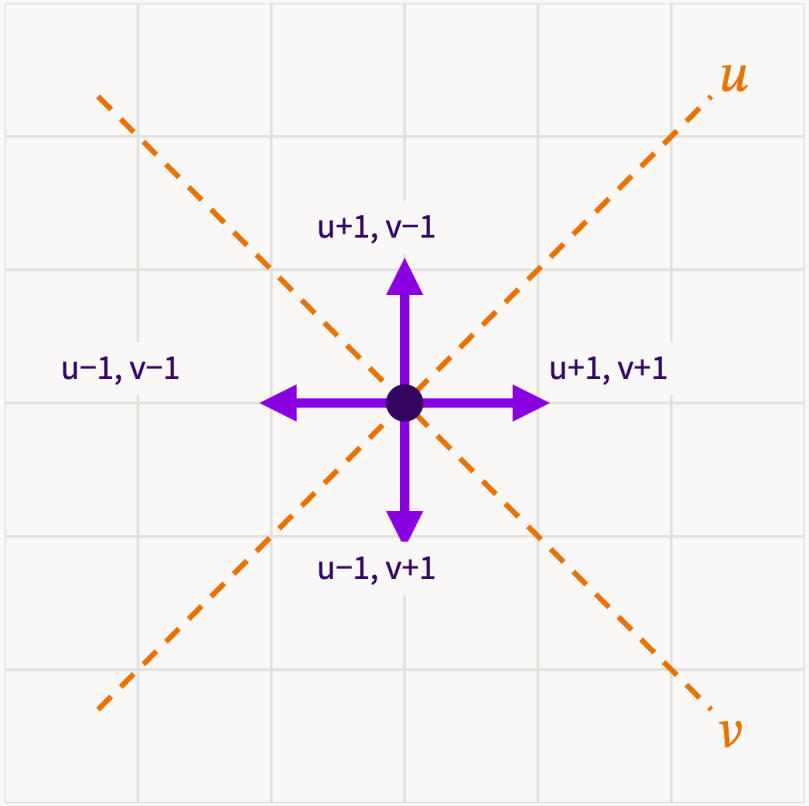
of 500 walkers, after 81,463 steps

One walker's trip home

step 91, still out

WHY? TWO WALKS IN ONE

THE 45° TRICK



Turn the grid by 45° and use the two diagonals, u and v, as new axes.

Every step moves one unit along u **and** one along v: each ±1, with chance ½, independently.

So a 2D walk is two independent 1D walks, and it is home only when both are home:

$$P(\text{home at step } 2n) = \left(\binom{2n}{n} \Big/ 4^n \right)^2 \approx \frac{1}{\pi n}$$

Add them up: (1/π) × the harmonic series. **Infinite**, so return is certain.

VOTE 3 OF 4 · SPACE

A walker in 3D space. Will it come back home with probability 1?

Each step: north, south, east, west, up or down, with chance ⅙ each.

Hands up: yes **Hands up: no**

VOTE 3 · THE ANSWER

NO: ONLY ABOUT 1 IN 3 RETURN

Left: one walker's 3D path, with x, y, z axes shown (orange dot = home). Right: 1,000 walkers. It stalls near a third.



Walkers back home at least once

33.4%

of 1,000 walkers, after 15,526 steps

WHY IS 3D DIFFERENT?

EACH DIMENSION ADDS A FACTOR

World	Chance home at step 2n	Add them up	The walker
1D · a line	$\approx \frac{1}{\sqrt{\pi n}}$	infinite	always comes home
2D · a grid	$\approx \frac{1}{\pi n}$	infinite (harmonic)	always comes home
3D · space	$\approx \frac{c}{n\sqrt{n}}$	**finite**	**may escape forever**

Why the 3D sum is finite: group the terms in blocks starting at 1, 2, 4, 8, … (the 3D rate is stated here, not proved; Pólya proved it in 1921).

$$2^k \times \frac{1}{2^k\sqrt{2^k}} = \left(\frac{1}{\sqrt{2}}\right)^k \qquad \sum_{k\geq 0}\left(\frac{1}{\sqrt{2}}\right)^k = \frac{1}{1-1/\sqrt{2}} \approx 3.41$$

PÓLYA'S THEOREM, 1921

On a line or a grid, a random walker always comes home. In space, it may wander off forever.

> *"A drunk man will find his way home, but a drunk bird may get lost forever."*
>
> Attributed to the mathematician Shizuo Kakutani

OUT IN THE WORLD

WHERE RANDOM WALKS LIVE

ATOMS

Pollen grains in water jiggle because molecules keep hitting them. Einstein's 1905 random-walk explanation of this "Brownian motion" helped convince scientists that atoms exist.

CELLS

Adam and Delbrück (1968) argued that molecules find their targets more efficiently by landing on a membrane and wandering in 2D than by searching in 3D.

THE WEB

Google's PageRank (1998) imagines a random surfer clicking links forever. The share of time it spends on each page ranks that page.

02

THE RANDOM VOTERS

Will a crowd of voters end up agreeing?

THE VOTER MODEL

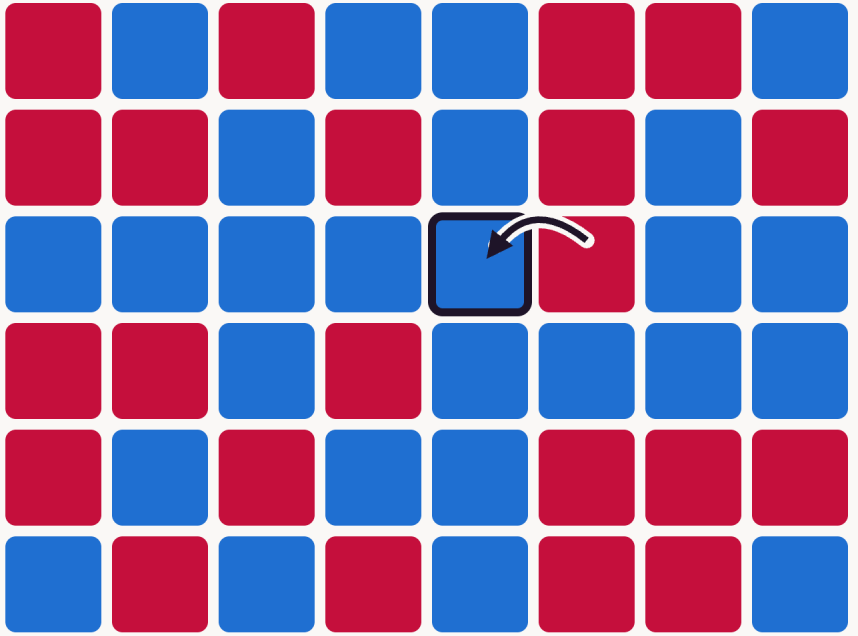
COPY A RANDOM NEIGHBOUR

Everyone holds an opinion: **red** or **blue**.

At random moments, a random person picks one of their neighbours at random and copies them.

No persuasion, no ideology: only imitation.

> Clifford and Sudbury (1973) introduced it for two species competing for territory. Holley and Liggett (1975) named it the voter model.



The outlined person picks a neighbour (the red one on the right) and turns red.

A FINITE GROUP

SMALL WORLDS ALWAYS AGREE

In any finite group, consensus is certain.

A TRAP

Once everyone agrees, copying changes nothing. All-red and all-blue are "absorbing" states of the Markov chain.

ALWAYS REACHABLE

From any starting colours, some run of copies leads to agreement. With finitely many states, the chain falls into the trap eventually.

A FAIR GAME

If everyone is connected to everyone, red's chance of winning equals red's current share, k/n. That is exactly gambler's ruin.

A FAIR GAME · THE MATHS

WHY RED'S CHANCE EQUALS ITS SHARE

1 · EVERYONE KNOWS EVERYONE, SO ONLY THE NUMBER OF REDS MATTERS

***k* goes up:** pick a blue person, who copies a red one

$$P(k \to k+1) = \frac{n-k}{n} \cdot \frac{k}{n-1}$$

***k* goes down:** pick a red person, who copies a blue one

$$P(k \to k-1) = \frac{k}{n} \cdot \frac{n-k}{n-1}$$

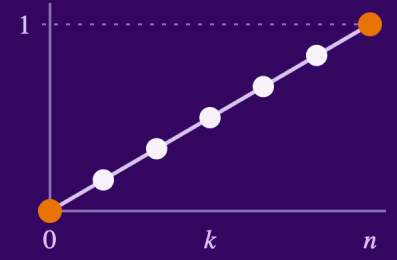
THE SAME TWO NUMBERS

So whenever *k* moves, up and down are equally likely. *k* does a **fair coin-flip walk** between 0 and *n*: exactly gambler's ruin.

2 · A FRESH START

Let *wk* be red's chance of winning from *k* reds. Skip copies that change nothing: the next real move is down or up, ½ each, then the game starts afresh.

$$w_k = \tfrac{1}{2}\, w_{k-1} + \tfrac{1}{2}\, w_{k+1}$$



Each value is the average of its neighbours, so all gaps are equal: a straight line from 0 up to 1.

$$w_k = \frac{k}{n}$$

VOTE 4 OF 4 · AN INFINITE WORLD

Infinitely many voters, starting 50–50 at random. Will every neighbourhood eventually agree?

On a line? | On a grid? | In 3D space?

VOTE 4 · THE ANSWER

AGREE IN 1D AND 2D, NOT IN 3D

Watch 1D and 2D hit 0% — full agreement — while 3D settles near 33% and stays there. The walker's pattern again. Coincidence?

1D · A LINE



Neighbours who disagree

8%

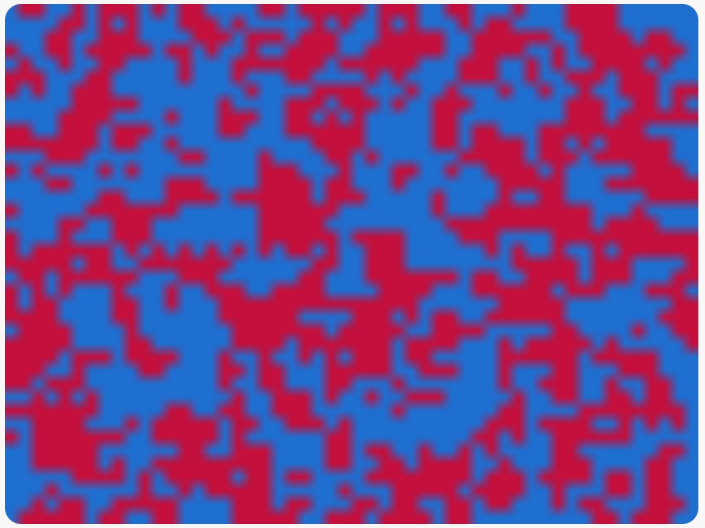
2D · A GRID



Neighbours who disagree

30%

3D · A SLICE

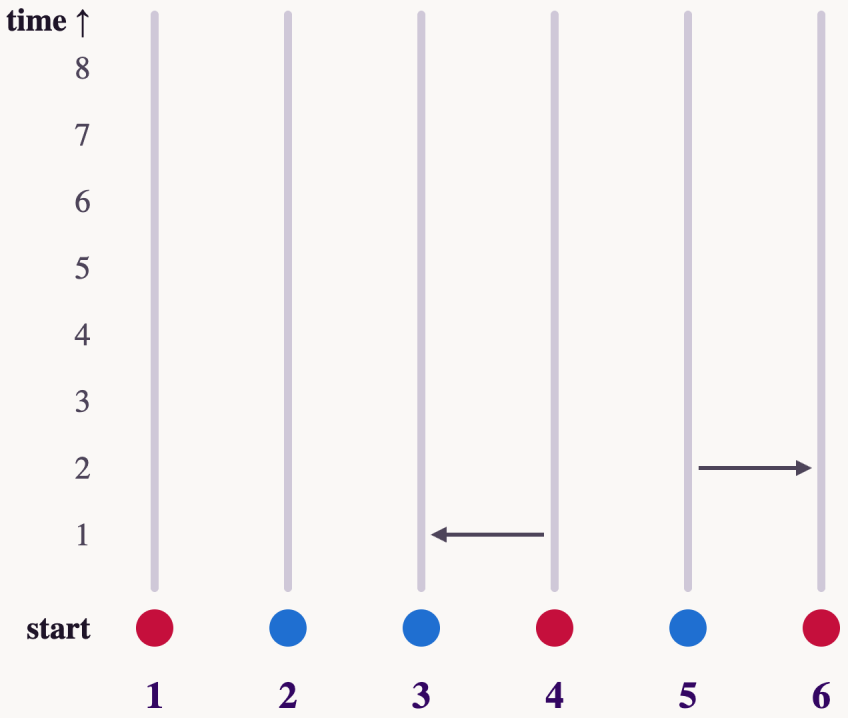


Neighbours who disagree

36%

THE REVEAL · BUILDING THE BOARD

LET'S BUILD THE BOARD



6 people in a row, each holding a red or blue opinion. A coin flip sets each starting colour, marked at the bottom.

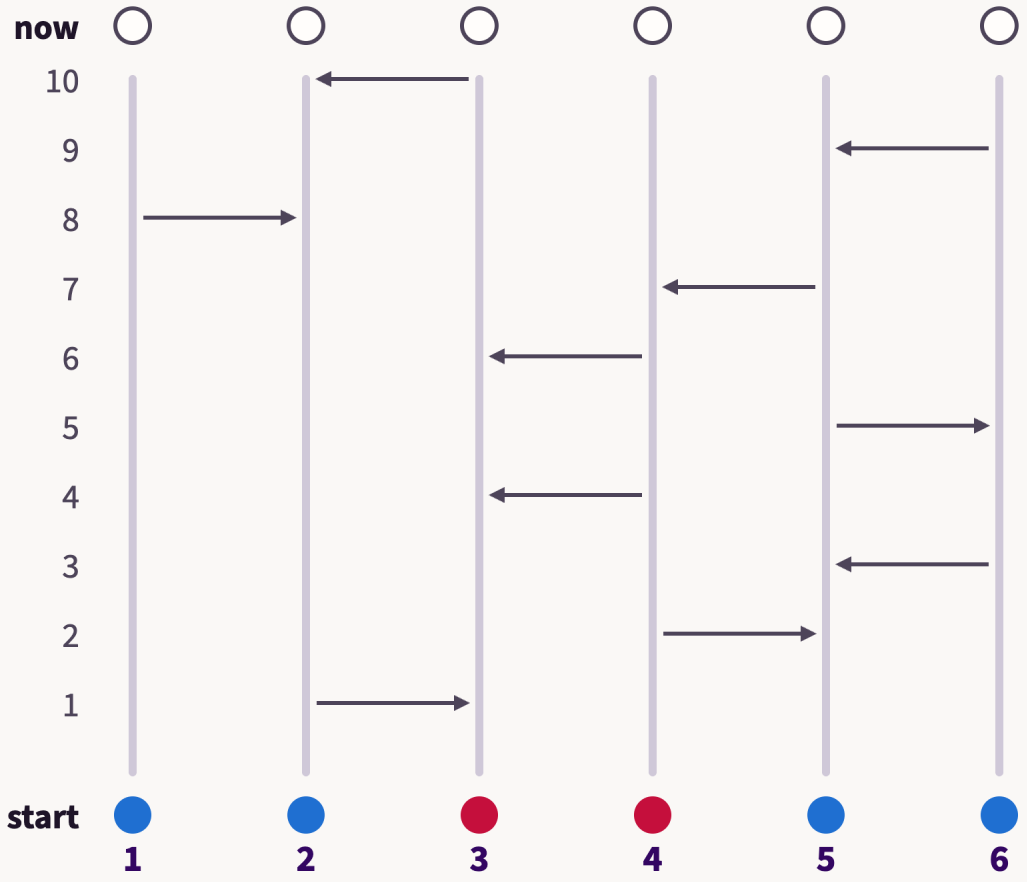
EACH ROUND

1. One person is picked at random to copy.
2. A coin flip decides: heads copies the left neighbour, tails the right. The two ends copy their only neighbour.
3. The copier's opinion changes to match.
4. An arrow is drawn from the person copied to the copier, **even if nothing changed**.

Each person's colour shows what they think now. The board shows how it happened.

THE REVEAL · 1 OF 4

READING THE BOARD



Columns are people 1 to 6. Time runs upwards, one row per round.

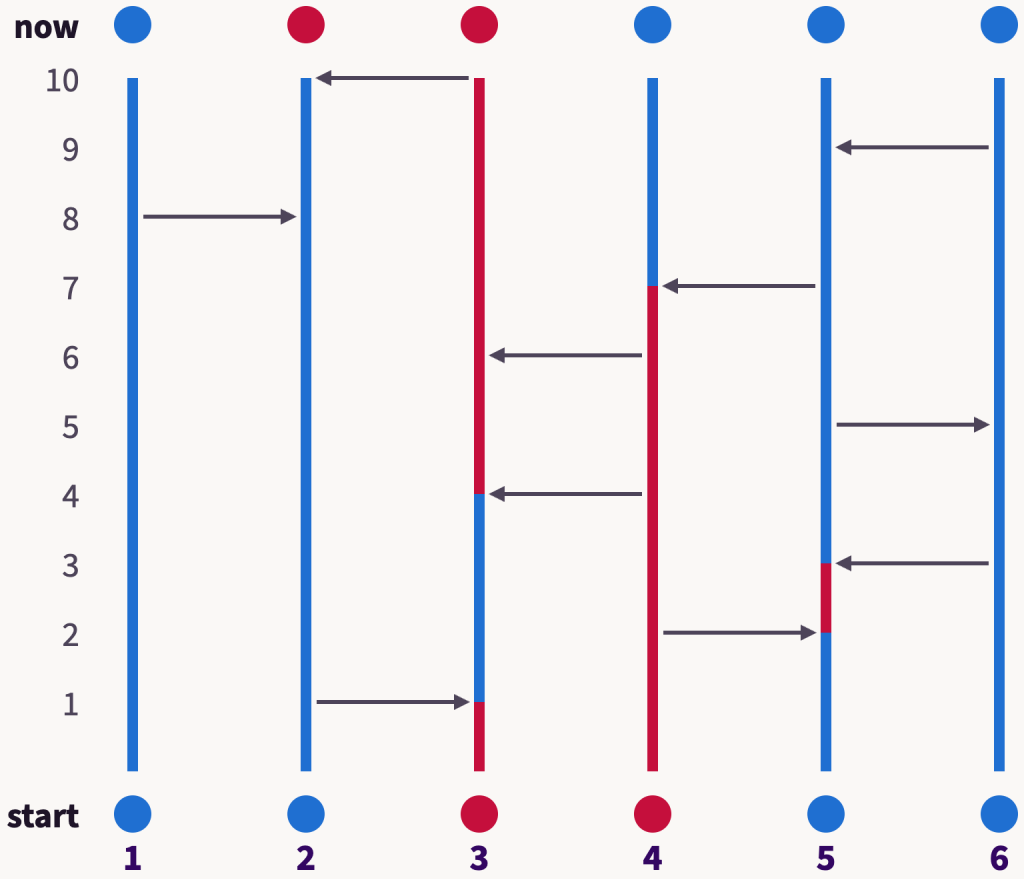
An arrow from column x into column y means "y copied x" in that round.

Bottom dots: the starting colours. Top circles: the colours now, still unknown.

The same kind of board as the previous slide: a clean example with 6 people and 10 rounds.

THE REVEAL · 2 OF 4

GOING FORWARDS



Running forwards, a colour travels up its column.

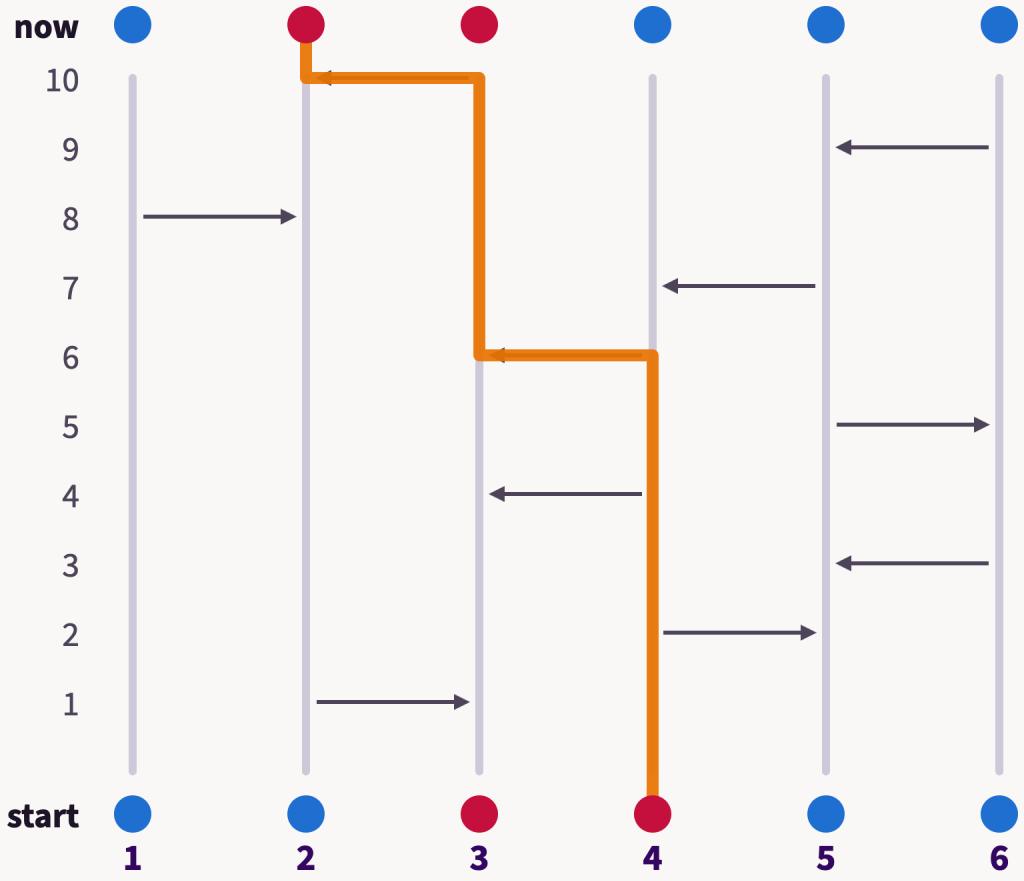
When an arrow points into a column, that person's colour is overwritten by the colour coming in.

"Silent" copies in rounds 5, 6, 8 and 9 change nobody's colour.

Now: **blue**, **red**, **red**, **blue**, **blue**, **blue**. But to get one person's colour this way, you have to follow everybody.

THE REVEAL · 3 OF 4

GOING BACKWARDS

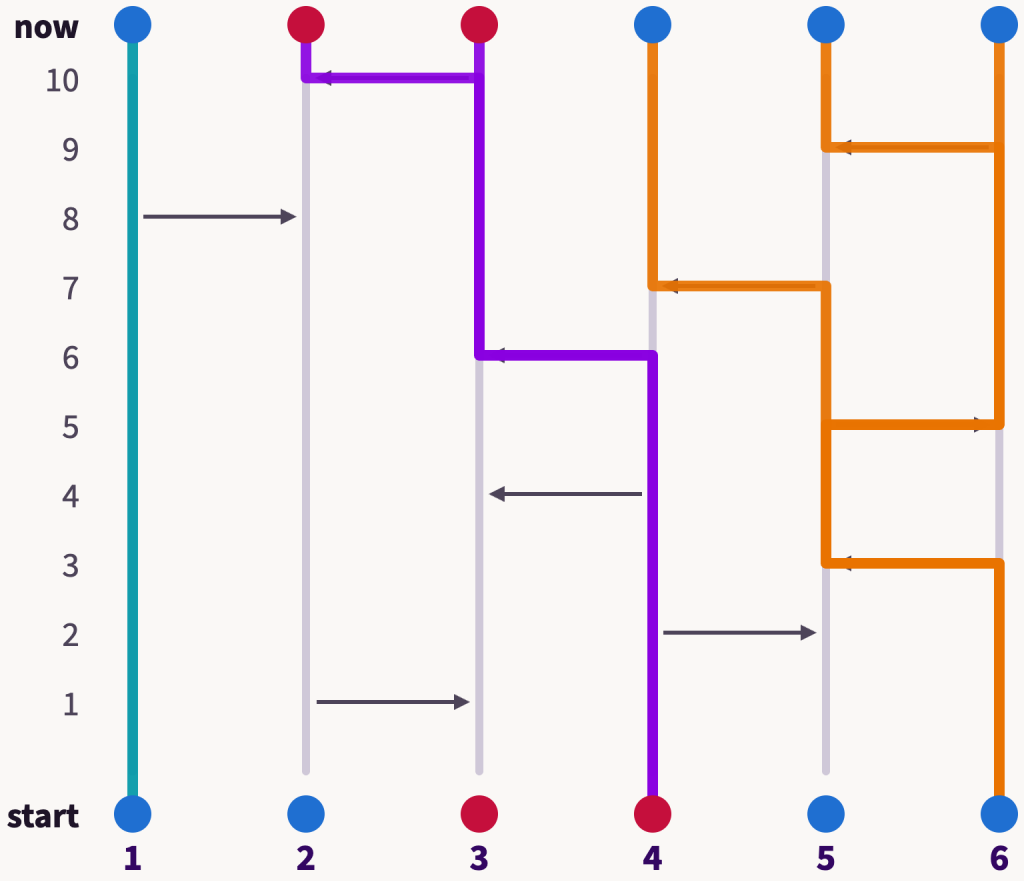


1. Put the pen at the top of one person's column and move down.
2. When an arrowhead points into the column, jump along the arrow to where it came from.
3. Ignore arrows leaving the column.
4. The dot at the bottom is the ancestor: your colour now is its starting colour.

Person 2: jump to 3 in round 10, then to 4 in round 6. Ancestor 4 started **red**, so person 2 is red now. Correct.

THE REVEAL · 4 OF 4

LINES MERGE



Once two lines meet, they follow the same arrows for ever: they **merge**.

- Person 1 → ancestor 1
- Persons 2, 3 → ancestor 4
- Persons 4, 5, 6 → ancestor 6

Six people, three ancestors. Merged people must agree. Unmerged people can disagree (3 and 4) or agree by luck (1 and 6).

Person 3's starting red has no descendants today: that opinion is lost.

THE KEY OBSERVATION

THE PAST IS A RANDOM WALK

Forward: **an opinion fans out like a tree.** Many people can copy you.

Backward: **you follow a single path.** You copied exactly one person, a uniformly random neighbour.

That's why looking backward is easier: a single path that keeps hopping to a random neighbour is a random walk.

Read an ancestry line downwards, one round at a time:

MOST ROUNDS

It stays put.

SOMETIMES

Its person was chosen to copy (chance 1 in n). Then it hops one step left or right, 50–50.

That is a random walk, running backwards in time.

THE PAPER FLIP

Turn the board upside down. The arrows are still independent left-or-right hops at random times.

So the past of the voters is a crowd of random walkers that merge when they meet: **coalescing random walks**.

In 2D or 3D, "left or right" becomes "one of this site's 4 (or 6) lattice neighbours" — the traced line is the same squiggly path as Part 1's walker.

THE PAYOFF

WHEN DO TWO PEOPLE DISAGREE?

$$P(\text{two people disagree now}) = 2p(1-p) \times P(\text{their ancestry lines have not met yet})$$

LINES HAVE MET

Same ancestor, so the two people agree.

LINES HAVE NOT MET

Different ancestors with independent starting colours, so they disagree with chance $p(1-p) + (1-p)p$.

p is the chance a person starts red. Starting 50–50, $2p(1-p) = ½$.

CLOSING THE LOOP

BACK TO PÓLYA

Two lines meet exactly when the gap between them closes. At most one line hops in any round, so **the gap itself does a random walk**. Agreement is Pólya's question in disguise.

1D · A LINE

The lines surely meet, so disagreement falls to 0. Every neighbourhood agrees.

2D · A GRID

The lines surely meet, slowly. Every neighbourhood agrees, eventually.

3D · SPACE

Neighbours' lines never meet with chance 66%. Starting 50–50: ½ × 0.66 ≈ **33% disagree for ever**.

That 33% is exactly where the 3D meter settled in the simulation.

HOLLEY AND LIGGETT, 1975

The two questions were the same question.

In 1D and 2D, any two people are sure to end up agreeing, just as the walker is sure to come home. In 3D, two neighbours can stay apart for ever, just as the walker can get lost for ever.

> *Forward, an opinion spreads through a crowd. Backward, it has a single history, and that history is a lost walker. So whether a crowd agrees comes down to whether a walker finds its way home.*

03

WHY IT MATTERS

Family trees, ecosystems, networks and open problems

GENETICS

FAMILY TREES COALESCE

Geneticists trace ancestry backwards in exactly this way. It's called **coalescent theory**.

Follow everyone's mother's mother's … line back in time: the lines merge, ending in one woman, "mitochondrial Eve" (Cann, Stoneking and Wilson, 1987).

Kimura's stepping-stone model (1953) has genes spreading between neighbouring villages, a cousin of the voter model. Neutral genes can take over by pure chance: **genetic drift**.

ASK THE ROOM

Does everyone here share a common ancestor?

Yes: trace far enough back and every line merges.

ECOLOGY AND SOCIETY

FORESTS, NETWORKS, STUBBORN VOTERS

FORESTS

Hubbell's neutral theory of biodiversity (2001): when a tree dies, a neighbour's seedling takes its place. A voter-style rule for whole ecosystems.

NETWORKS

Opinion dynamics on social networks: who copies whom, and how fast a network drifts towards agreement.

STUBBORN VOTERS

People who never change their minds (researchers call them zealots). With some on each side, consensus can be stopped altogether (Mobilia, 2003).

THE BIGGER PICTURE

MARKOV CHAINS EVERYWHERE

7

SHUFFLING

riffle shuffles mix a 52-card deck (Bayer and Diaconis, 1992).

MONTE CARLO

Random walks that simulate atoms (Metropolis, 1953), and today weather, finance and AI.

SEARCH

PageRank: a random surfer's time on each page ranks the web.

MONOPOLY

Dice moves make a Markov chain, which tells you which squares get landed on most.

Everything tonight was a Markov chain: the walker's state is its position; the voters' state is the whole map of opinions.

STILL BEING DISCOVERED

THIS IS CURRENT RESEARCH

Randomness on lattices like these has won Fields Medals, the top prize in mathematics:

2006	Wendelin Werner
2010	Stanislav Smirnov
2022	Hugo Duminil-Copin

AN OPEN PROBLEM ANYONE CAN STATE

A **self-avoiding walk** never visits the same square twice. How far from home does it typically get after n steps?

Even on a flat grid, nobody has proved the answer.

BACK TO PÓLYA'S WOOD

THREE TAKEAWAYS

A wood is two-dimensional, so Pólya was bound to meet that couple again. Birds are not so lucky.

1

Dimension changes everything.

In the model, what matters isn't how many neighbours you have, but how fast the crowd around you grows as you look further away. On a line or a flat grid, any two people end up agreeing; in 3D, disagreement lasts for ever.

2

Looking backwards can make a hard problem easy.

3

Simple random rules model real life, and the maths is still being discovered.

BEFORE WE FINISH

A MATHEMATICIAN'S RIGOUR

A maths bachelor's student, a PhD student and a professor are on a train to a conference. Out of the window, they see a black cow.

BACHELOR'S STUDENT	"Cows in this country are black."
PHD STUDENT	"At least one cow in this country is black."
PROFESSOR	**"At least one side of at least one cow in this country is black."**

IF YOU WANT MORE

FURTHER READING

Doyle and Snell, *Random Walks and Electric Networks*. Free online, and readable for sixth-formers.

Pal and Mesikepp, *Finite Markov Chains and Monte-Carlo Methods* (2025). Undergraduate notes.

Leonardo Rolla's online simulations of the voter model and friends.

Levin, Peres and Wilmer, *Markov Chains and Mixing Times*. Advanced.

THANK YOU FOR YOUR ATTENTION

Questions? Find me on LinkedIn.

linkedin.com/in/catalin-botezat

BACKUP

Slides for questions

BACKUP · A SECOND PROOF FOR 1D

1D RETURN VIA GAMBLER'S RUIN

$$P(\text{reach } n \text{ before } 0, \text{ starting at } k) = k / n$$

WHY

Each chance is the average of its two neighbours' chances, so they lie on a straight line from 0 (at 0) to 1 (at n). Pal and Mesikepp, Theorem 2.1.

SO

After its first step the walker is at 1. To escape for ever it must reach n before 0 for every n: chance at most 1/n for every n, which is 0. It returns.

BACKUP · FINITE NETWORKS

WHO WINS ON A FINITE NETWORK?

On any connected network, red's chance of winning is the **degree-weighted** share of red supporters:

$$\mathrm{P}(\text{red wins}) = (\text{sum of the degrees of red people}) \div (\text{sum of all degrees})$$

A person's degree is their number of neighbours. These weights are exactly the walker's long-run share of time at each person (Pal and Mesikepp, Theorem 1.4).

If everyone is connected to everyone, all degrees are equal, so it's just red's share k/n: gambler's ruin again.

BACKUP · MORE ABOUT VOTERS

BORDERS, BIAS AND SPEED

1D BORDERS WALK

The borders between red and blue blocks move as random walkers, and two borders that meet annihilate (Toninelli, §3.6).

BIASED VOTERS

If one type spreads faster, like a fitter gene, it can take over even from a single individual (Toninelli, §1.7).

WHY 2D IS SLOW

In 2D, disagreement between neighbours fades only about as fast as 1 / log(time). That's why the 2D simulation needs a small grid to finish live.

BACKUP · MORE ABOUT VOTERS

DOES IT EVER SETTLE DOWN?

NOT BY A DEADLINE

"Eventually" has no fixed round number. In 1D/2D, two neighbours meet with certainty only if you're willing to wait arbitrarily long — fast in 1D, astronomically slow in 2D.

ANY STARTING MIX

Whether lines meet is pure lattice geometry, not colours. Change the starting proportions and the 1D/2D-versus-3D split is unchanged; only the *amount* of disagreement rescales.

TWO EQUILIBRIA

1D/2D only have "boring" stable states: everyone red, or everyone blue. 3D has a genuinely new one, where both colours coexist forever in a fixed statistical mix.

CLUSTERING (1D, 2D)

Pick any two people, however far apart: with probability 1 they eventually agree and never flip again. Zoomed out, patches of one colour keep growing without any colour ever winning everywhere.

COEXISTENCE (3D AND UP)

The fraction of disagreeing neighbours settles down and stops changing — that levelling-off meter reading IS the stable state. Both colours survive forever, side by side.

BACKUP · SAME PÓLYA, DIFFERENT THEOREM

THE PÓLYA URN

Start with one red ball and one blue ball. Draw a ball, then put it back with another of the same colour. Repeat. The rich get richer.

A SURPRISE

After n draws, the number of red balls is equally likely to be any of 1, 2, …, n + 1 (Pal and Mesikepp, Theorem 2.7).

IN THE LONG RUN

The share of red settles down, at a uniformly random value between 0 and 1. Early luck locks in (Theorem 2.8).

BACKUP · THE RIGOROUS VERSION

WHERE C/(N√N) REALLY COMES FROM

The exact return probability on $\mathbb{Z}$^d has a closed form via Fourier inversion — no rotation trick needed, and it works in every dimension at once:

$$P(S_m = 0) = \frac{1}{(2\pi)^d} \int_{[-\pi,\pi]^d} \varphi(\theta)^m \, d\theta, \qquad \varphi(\theta) = \frac{1}{d} \sum_{j=1}^{d} \cos\theta_j$$

ϕ is the characteristic function of one step. Near θ = 0, cos θ ⱼ ≈ 1 − θ ⱼ ²/2, so ϕ(θ)^m behaves like a Gaussian; everywhere else |ϕ(θ)| < 1, so those parts vanish exponentially fast. The Gaussian integral gives the local central limit theorem:

$$P(S_m = 0) \sim 2\left(\frac{d}{2\pi m}\right)^{d/2}$$

d = 1
2(1/4πn)^½ = 1/√(πn) — matches the exact binomial value.

d = 2
2(2/4πn)¹ = 1/(πn) — matches the 45° rotation result exactly.

d = 3
the same formula now gives the constant directly:

$$P(\text{home at } 2n) \sim 2\left(\frac{3}{4\pi n}\right)^{3/2} \approx \frac{0.233}{n\sqrt{n}}$$